A Framework for the Estimation of Uncertainties and Spectral Error Correlation in Sentinel-2 Level-2A Data Products

Javier Gorroño, Luis Guanter, Lukas Valentin Graf, and Ferran Gascon

***Abstract*— The Copernicus Sentinel-2 (S2) satellite mission acquires high spatial resolution optical imagery over land and coastal areas. Delivering uncertainty estimates and spectral error correlation alongside S2 data products facilitates the constrain of retrieval algorithms, propagates further downstream the retrieval uncertainty, and, finally, makes informed decisions to end-users. This study presents a framework to produce uncertainty estimates and spectral error correlation associated with the S2 L2A data products (i.e., surface reflectance). This framework has been implemented in a prototype code available at https://doi.org/10.5281/zenodo.11971517. The uncertainty considers both the Level-1 (L1) uncertainty estimates for the top-of-atmosphere (TOA) reflectance factor and the atmospheric correction. The L2A error distribution cannot be systematically described as a normal distribution; the transformation can be nonlinear and without an explicit mathematical model. Thus, a multivariate Monte Carlo model (MCM) rather than the law of propagation of uncertainty (LPU) is selected for uncertainty propagation. We show results for surface reflectance uncertainty over the Amazon forest and Libya4 desert site. It illustrates the large uncertainty and spectral error correlation variations depending on the scene. The comparison of a multivariate MCM against an LPU propagation methodology indicates the limitations of the latter for scenes dominated by the atmospheric path. Its implementation as an operational per-pixel processing and dissemination of both the uncertainty and spectral error correlation becomes challenging. Therefore, this methodology is not expected to run at an operational level but serves as the basis to define a strategy for an operational one.**

***Index Terms*— Copernicus, Level-2A, spectral error correlation, surface reflectance, uncertainty.**

I. Introduction

EARTH Observation (EO) via satellite remote sensing has exponentially increased in terms of data and applications in the last decade. It is nowadays an important source of information about the Earth system with areas of applications that include, for example, climate studies or agriculture monitoring. The increased complexity of these applications and the constantly increasing number of data products have triggered the need to include a quality indicator that describes the compatibility between satellite products and their suitability for particular applications. Some initiatives such as the Quality Assurance framework for EO (QA4EO) or Fidelity and uncertainty in climate data records from EOs (FIDUCEOs) actively work on a high-level framework that becomes the basis of a rigorous quality assessment for EO satellite missions [2], [3].

Recent efforts by the community have made it possible for some satellite missions to offer operational Level-1 (L1) uncertainty estimates at top-of-atmosphere (TOA) radiance/reflectance factor. The Moderate Resolution Imaging Spectroradiometer (MODIS) instrument onboard the Aqua and Terra missions produces uncertainty estimates associated with the L1B product [4], [5], and the Copernicus Sentinel-2 (S2) satellite mission does it for the L1C products through the L1 Radiometric Uncertainty Tool (L1-RUT) [6]. The Copernicus Sentinel-3 mission also delivers detailed uncertainty information for the L1B products. They are included at a pixel level for both the Sea and Land Surface Temperature Radiometer (SLSTR) instrument [7] and the Ocean and Land Color Instrument (OLCI) [8] instruments.

The Copernicus S2 satellite mission currently comprises two satellites (S2A and S2B) and provides continuous monitoring of terrestrial surfaces and coastal waters at a global scale with better than a five-day revisit [9]. It carries the Multi-Spectral Instrument (MSI) with 13 spectral bands in the visible and near-infrared (VNIR) and the shortwave infrared (SWIR) at spatial resolutions from 10 to 60 m. The S2 Level-2A (L2A) data product is generated after the atmospheric correction of the L1C observations. The L2A product data approximate to the hemispherical–directional reflectance at a surface (or top of canopy level) [10], [11]. From here on, we refer to it as *surface reflectance* for simplification. The L2A data products are delivered in a set of orthoimages in Universal Transverse Mercator (UTM) projection of about 100 × 100 km [12]. The L2A product represents the most advanced data offered to the users, and it is the input to many different land surface parameters such as leaf area index (LAI), the fractional vegetation cover (FVC), and the fraction of absorbed photosynthetically active radiation (FAPAR) in [13]. Including uncertainty information associated with these data products would be highly relevant to improve retrieval algorithms or provide uncertainty information in different end-user applications such as phenology metrics.

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Several efforts have been made in bringing uncertainties into land surface parameter retrievals, for example, in [14] where an uncertainty estimate is associated with the LAI, FVC, and FAPAR. The uncertainty is quantified as a combination of the predictive standard deviation and the propagation of the Bi-Directional Reflectance Function (BRDF) model uncertainty. In addition, the C3S LAI and FAPAR v4 products [15] include the possibility to manage the whole covariance information at the input Bi-Hemispheric Reflectance (BHR) and propagate to the output results (LAI and FAPAR) [16].

Although these examples represent a clear advance in providing uncertainty estimates in a rigorous manner for land surface parameters, limited detailed uncertainty and covariance information in the observations hinders further efforts. Indeed, it illustrates how the absence of operational uncertainty estimates for surface reflectance products limits its propagation to further processing levels mainly in land retrieval methods. Thus, the main objective of this project is the generation of uncertainty and spectral error correlation information (i.e., full covariance matrix) associated with the S2 L2A data products.

The impact of this novel information is not only addressed to land surface parameters. It is also expected (but not limited to) to perform the following.

1) *Derive New and Better-Quality Metrics for End-User Applications:* For example, in the field of agricultural monitoring, the study in [17] explores an uncertainty propagation framework for the S2 data products that translates the L1C and L2A uncertainty into an uncertainty in the vegetation parameters and ultimately into land surface phenological metrics.
2) *Constrain Measurement Space in Retrieval Methods:* For example, in optimal estimation methods, the accurate definition of the observation covariance matrix can better constrain the predictions [18].
3) *Improve the Compliance and Quality Information of L2A Data Products:* Reliable information of quantitative EO data products is crucial for decision-making processes such as regulatory initiatives or contractual negotiations for the operational exploitation of EO data [19].

The article briefly defines the atmospheric correction of the L2A data products in Section II-A, and then, it describes each one of the considered uncertainty contributions in Section II-B. The multivariate Monte Carlo method (MCM) is synthesized in Section II-C. An alternative hybrid version of the law of propagation of uncertainty (LPU) has been implemented and is described in Section II-D. The results in Section III include examples for a scene of the Amazon forest and another in the Sahara desert in Section III-A. Finally, the difference between an LPU approach and an MCM method is presented in Section III-B.

II. Methodology

A. Atmospheric Correction in Sen2Cor

The current L2A operational products delivered by the European Space Agency (ESA) are generated with Sen2Cor both from the user side and from the ground segment [20]. Sen2Cor performs a preprocessing of L1C reflectance including a scene classification, an atmospheric correction, and a conversion into surface reflectance orthoimages that define the main content of the L2A product. In addition, the algorithm also delivers an aerosol optical thickness (AOT) map, a water vapor (WV) map, and a scene classification map.

The atmospheric correction is based on precompiled lookup tables (LUTs) that contain the main atmospheric functions for a large homogeneous Lambertian surface generated with the Library for Radiative Transfer (LibRadtran) [21], [22].

The input for the Sen2Cor atmospheric correction is the TOA radiance L_{TOA} that is converted from the L1C TOA reflectance ρ_{TOA} for a pixel in the column i and row j as follows:

$$L_{\mathrm{TOA}}(i,j) = \frac{E_S \cdot \cos\Theta_s}{\pi \cdot d^2}\rho_{\mathrm{TOA}}(i,j) \tag{1}$$

where Θ_s refers to the sun zenith angle (SZA), d refers to the Sun–Earth distance in astronomical units, and E_S refers to the sun irradiance that, for the S2 data products, is modeled based on the Thuillier model [23].

The L2A atmospheric correction algorithm is based on ATCOR [24]. Over a flat terrain it neglects, in a first step, the influence of the neighborhood (adjacency effect) and the surface reflectance is obtained as

$$\rho^{\mathrm{I}}(i,j) = \frac{\pi \cdot (L_{\mathrm{TOA}}(i,j) - L_p)}{\tau \cdot E_g} \tag{2}$$

where τ is the ground-to-sensor transmittance, L_p refers to the path radiance (i.e., solar radiation reflected back to space by the atmosphere before reaching the surface), and E_g refers to the downwelling irradiance on the ground.

In the second step, the initial surface reflectance is corrected for the adjacency effect following the expression:

$$\rho^{\mathrm{II}}(i,j) = \rho^{\mathrm{I}}(i,j) + q \cdot (\overline{\rho^{\mathrm{I}}} - \rho(i,j)) \tag{3}$$

where $\overline{\rho^{\mathrm{I}}}$ represents a mean reflectance of the pixels over a selected range of the adjacency effect and $q = (\tau_{\mathrm{diff}}/\tau_{\mathrm{dir}})$ represents the strength of the adjacency effect. It is the ratio of the diffuse to direct ground-to-sensor transmittance.

The spherical albedo of the atmosphere s on the global flux of the surface needs to be considered. The initial surface reflectance is set to 0.15 for all the spectral bands and needs to be adapted to the scene-specific level [24]. Sen2Cor does not include this correction because it is generally negligible. However, we have modeled this step in the prototype implementation so that it can be tested for specific scenarios where it might have a larger impact. The correction is written as

$$\rho^{\mathrm{III}}(i,j) = \rho^{\mathrm{II}}(i,j) \cdot [1 - (\overline{\rho^{\mathrm{I}}} - \rho_r) \cdot s] \tag{4}$$

where ρ_r is the reference reflectance set at a value of 0.15.

B. L2A Product Uncertainty: Tree Diagram and Uncertainty Sources

The first step in our analysis involves the definition of the main processing steps and equations for the generation of L2A products (see Section II-A).

From here, we can identify the potential sources of uncertainty that stems from each one of the processing steps. The

result of this analysis can be synthesized in an uncertainty tree diagram [3] and is presented in Fig. 1.

The tree diagram includes the processing from the L1C products and the different steps presented in Section II-A. Here, the term $u(0)$ represents assumptions and approximations that have been identified and are not evaluated. The rest of the identified sources of uncertainties are described here.

1) L1C Uncertainty: The L1C uncertainty is described in [6] and includes several contributions such as the diffuser calibration uncertainty or the dark signal stability.

The original code (available online at https://github.com/senbox-org/snap-rut) has been converted into a sampling scheme based on an MCM methodology. The code considers the spectral error correlation between the considered L1C uncertainty contributions based on the description in [25]. The result is a set of samples for each band that represents the error distribution at L1C reflectance and inherently includes the spectral error correlation between them.

The irradiance uncertainty has been also included in the code but only for test purposes. This contribution is canceled out when considering TOA reflectance uncertainty [6]. However, the data in Sen2Cor are converted back into TOA radiance [see (1)], and although it is canceled out again during the atmospheric correction, it is possible that a small effect remains and/or other effects (e.g., rounding) need to be considered. Uncertainty for this contribution has been modeled based on estimates in VNIR from [26] and SWIR from [23]. Correlation information has been based on the assumption that the error correlation decreases with the spectral distance.

2) Uncertainty in Atmospheric Parameters (RTM Input Uncertainty): The input to the radiative transfer code is a set of parameters (e.g., molecular profile, trace gases, and aerosol distribution) that define the atmospheric model. Then, this model is converted into a set of optical properties (optical depth, single scattering albedo, phase function, and boundary conditions) that solve the radiative transfer equation [21], [22]. The specific solver is that of Discrete-Ordinate-Method Radiative Transfer (DISORT) [27], and the full model description is presented in [28]. In the implementation presented here (see Sections II-C and II-D), we opted for a multivariate Monte Carlo propagation through DISORT. Despite a detailed mathematical model, it becomes very complex to consider mathematical approximations, rounding values, or internal convolution and regridding in Libradtran. Thus, uncertainty on the input atmospheric parameterization is propagated through the solver and results in uncertainty on the different atmospheric functions τ, L_p, E_g, and s (see Section II-A).

Here, we detail each one of the main parameters that are included and how they have been parameterized together with the associated uncertainty estimate.

1) *Aerosol Optical Thickness:* Calculated based on the dark dense vegetation (DDV) algorithm in [29] and AOT meteorological estimates from European Centre for Medium-Range Weather Forecasts (ECMWF) CAMS data as a fallback solution [30]. The theoretical uncertainty for the former becomes highly challenging (e.g., assumed ratios for the bands and correlation). Thus, for this prototype version, the values of uncertainty have been obtained from a diagnostic uncertainty based on the mission requirement and the AOT independent validation. On the one hand, the uncertainty requirement for the AOT is modeled as

$$\sigma_{\mathrm{AOT}}^{\mathrm{requirement}} = 0.1 \cdot \mathrm{aot} + 0.03. \tag{5}$$

On the other hand, the extensive validation against AERONET stations worldwide has found systematic underestimation of the AOT value for both DDV and CAMS estimates [31]. Following the recommendations of the GUM [32] for uncorrected systematic errors, we have modeled the total AOT uncertainty as a sum of the requirement uncertainty $\sigma_{\mathrm{AOT}}^{\mathrm{requirement}}$ and the systematic error obtained from the validation as follows:

$$\sigma_{\mathrm{AOT}}^{\mathrm{DDV}} = \sigma_{\mathrm{AOT}}^{\mathrm{requirement}} + |-0.56 \cdot \mathrm{aot} + 0.07| \tag{6}$$

$$\sigma_{\mathrm{AOT}}^{\mathrm{CAMS}} = \sigma_{\mathrm{AOT}}^{\mathrm{requirement}} + |-0.46 \cdot \mathrm{aot} + 0.09|. \tag{7}$$

2) *Water Vapor:* This is based on the APDA (atmospheric precorrected differential absorption) algorithm [33]. The same reasoning as for the AOT case applies here, and the assigned uncertainty is based again on the requirement values expanded with the systematic error obtained from the AERONET validation as follows:

$$\sigma_{\mathrm{WV}}^{\mathrm{requirement}} = 0.1 \cdot \mathrm{wv} + 0.2 \tag{8}$$

$$\sigma_{\mathrm{WV}}^{\mathrm{APDA}} = \sigma_{\mathrm{WV}}^{\mathrm{requirement}} + |-0.1 \cdot \mathrm{wv} + 0.03|. \tag{9}$$

3) *Ozone:* It is set to 3% based on a comparison from total column ozone (TCO) satellite estimations against ECMWF ERA-interim (currently superseded by ECMWF ERA-5) in [34].

4) *SZA, RAA, and VZA:* Uncertainty assumed negligible compared to other sources of uncertainty (e.g., LUT sampling).

5) *CH_4, CO_2, and Altitude:* Assigned a mean of 1.8 and 400 ppm and mean tile value from the metadata, respectively; standard deviation allocations of 0.1 and 40 ppm and 10%, respectively.

3) Adjacency Correction Uncertainty [$u(\overline{\rho^{\mathrm{I}}})$]: The adjacency effect is produced by radiation coming from neighborhood pixels and scattered into the line-of-sight of the sensor. The standard adjacency kernel window size is 2 km. This adjacency range is not a critical parameter for most pixels but for areas with a high surface contrast (e.g., a lake surrounded by forests) [20]. However, the simplified modeling of this parameter reduces the processing time but might not capture the spatial dependence of the adjacency effect. For example, more complex definitions as in ATCOR include a range-dependent weighting [24].

For this first implementation of the L2A budget, the term $\overline{\rho^{\mathrm{I}}}$ in (3) has been modeled with an uncertainty of 3% and a spectral error correlation based on the spectral distance.

4) Lambertian Assumption Error [u(BRDF)]: The retrieval presented in Section II-A assumes a Lambertian surface. However, in a real scenario, the target surface is expected to be anisotropic, and this results in an error in the estimated

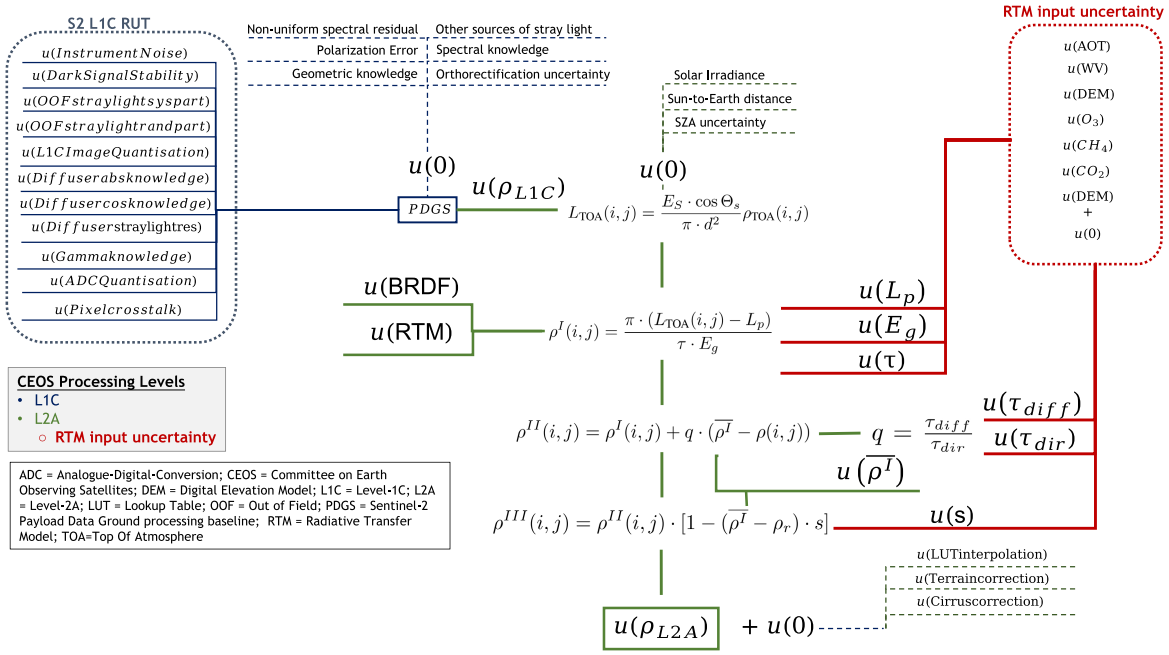


Fig. 1. Uncertainty tree diagram for the S2 L2A products considering the flat terrain case.

surface reflectance. These errors depend on the specific target and spectral band. For example, relative errors due to the Lambertian assumption on the surface reflectance have been calculated as 3%–12% in the visible and 0.7%–5.0% in the near-infrared in [35]. In another example in [36], mean errors were 3%–7% in the red and 2%–5% in the near-infrared (depending on optical depth).

For this implementation, we set an uncertainty of 3% for all bands. It might be slightly optimistic for some visible bands but average case in NIR bands.

5) LibRadtran Uncertainty u(RTM): This last contribution represents the uncertainty introduced by the radiative transfer itself. Benchmarking of radiative transfer typically relies on the comparison of ideal cases such as monochromatic radiation or Lambertian surface assumptions [37]. Although this might be a good initial assessment to validate and flag issues between radiative transfer codes, it does not represent real conditions for EO satellite observations.

Recent work in [38] considers pseudoinvariant sites to infer uncertainty in radiative transfers with a data-driven method. The study in [39] also presents specific radiative transfer model (RTM) disagreements over pseudocalibration sites. This second study provides detailed results for different codes (among them Libradtran) and for the Sentinel 2 satellite mission. These per-band errors are calculated as the maximum disagreements between mean bias errors over a long temporal trend of S2 satellite observation against simulated scenes over the Libya-4 calibration site. Although it covers a single site, they are the most detailed and S2-specific estimates of inherent Libradtran uncertainty, and we have included them in the first version of the L2A uncertainty. In the VNIR bands, the estimated errors are from 1% to 2%. The largest relative error is reported for B11 and B12 at the 2%–3% where the absorption parameterization (in this case the REPTRAN model available for S2 in Libradtran [40]) has limited capability to reproduce the varying molecular absorption. The error for B8 is reported above 3%, but, here, we will reduce the error to 1.2% since a large contribution was due to the limited spectral resolution of 6SV (see [41]) that is not present in Libradtran.

C. Multivariate Monte Carlo Methodology

The most common approach for uncertainty estimation is referred to as guide to the expression of uncertainty in measurement (GUM) and involves the analytical calculation of the uncertainty through partial derivatives and correlation coefficient [32], that is, the LPU. The LPU was the selected methodology for the S2 L1C uncertainty since the expected radiometric transformations are largely linear, and a normal error distribution can be assumed for the output measurand [6].

The L2A data products from S2 are generated with Sen2Cor processor [42]. It is based on the retrieval of the surface reflectance based on atmospheric functions precalculated with Libradtran.

The RTM converts the parameterization of the atmosphere model (e.g., AOT and height) into an optical model that solves the radiative transfer equation. Although the RTM can be defined by a set of equations [21], its mathematical description is limited. For example, the majority of radiative transfer codes include numerical integration and other approximations, which results in a very complex mathematical formulation [43]. When considering an entire radiative transfer code, this becomes even more complex since it might involve an internal convolution of the atmospheric transmittance or other effects such as rounding values [28].

In general, we find no explicit mathematical model of the atmospheric functions and no knowledge of their distribution shape, dependence, or linearity in the model. In that scenario, the MCM is the selected choice since it can work without

an explicit mathematical model and can model the input and output dependence and distribution shape [44].

The Monte Carlo method is based on three different steps.

1) The assignment of probability distributions to the input quantities as defined in Section II-B. All inputs, including L1C data, are modeled with a spectral error correlation matrix and an uncertainty per band.
2) We extract samples from the input quantities and calculate the processing steps, as defined in Section II-A. It involves the run of Libradtran to generate the error distribution associated with each one of the atmospheric functions [21], [22] and the calculation of (2)–(4) to obtain the surface reflectance.
3) The previous steps are repeated in an iterative manner. The output represents an error distribution (with implicit error correlation between S2 bands) where the measurand (here the surface reflectance) is contained [45]. Here, we offer the uncertainty ($k = 1$) based on the standard deviation of the surface reflectance samples in a normal distribution. If the distribution is considerably asymmetric, [45] recommends the use of the shortest coverage interval that corresponds to a specific coverage factor (e.g., 68.27% for $k = 1$).

This method has been implemented in a software version and is accessible in [1].

D. Law of Propagation of Uncertainty

As mentioned previously, it is rather complex to model a mathematical expression of the RTM. However, the combination of the atmospheric functions has been described in (2)–(4). Thus, here, we propose an alternative method that combines an MCM method to estimate the probability distribution of the atmospheric functions and the LPU methodology for their combination.

The same software in [1] provides the distribution for each one of the atmospheric functions L_p, E_g, τ, and s. Then, a corresponding value of uncertainty can be obtained for each atmospheric function and the correlation among them. Furthermore, the sensitivity coefficient for each atmospheric function can be defined from the defined (2)–(4).

In mathematical terms, the LPU framework is defined in matrix form as

$$U = C \times V \times R \times V^T \times C^T \tag{10}$$

where C represents the diagonal matrix of the sensitivity coefficients where the nonzero entries are defined as

$$\mathbf{C}[i,i] = \left[\frac{\partial \rho_{\mathrm{L2A}}}{\partial L_{\mathrm{TOA}}} \quad -\frac{\partial \rho_{\mathrm{L2A}}}{\partial L_{\mathrm{TOA}}} \quad \frac{\partial \rho_{\mathrm{L2A}}}{\partial \tau} \quad \frac{\partial \rho_{\mathrm{L2A}}}{\partial E_g} \quad \frac{\partial \rho_{\mathrm{L2A}}}{\partial s} \right] \tag{11}$$

and V is a diagonal matrix with the nonzero entries defining the individual uncertainty sources corresponding to the atmospheric functions and the L1C radiance as follows:

$$\mathbf{V}[i,i] = \left[\sigma(L_{\mathrm{TOA}}) \quad \sigma(L_{\mathrm{path}}) \quad \sigma(\tau) \quad \sigma(E_g) \quad \sigma(s) \right]. \tag{12}$$

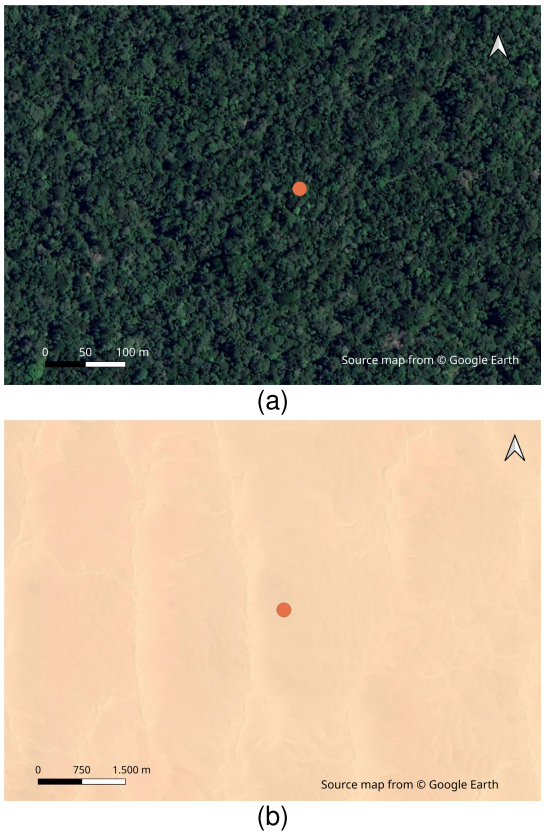


Fig. 2. Selected sites with (a) containing the Amazon forest centered at the latitude and longitude −6.2869°, −66.8652° and (b) containing the Libya4 desert at the latitude and longitude 28.55°, 23.39°.

The R matrix is the error correlation matrix between the individual uncertainty sources. On top of this uncertainty, an effect related to adjacency or other effects might be added.

This analytical and matrix implementation of the uncertainty is an efficient method for integration in the Sen2Cor processor since it minimizes the computational requirements on the user side by precompiling the uncertainty values for each atmospheric function and correlation. These values could be stored alongside each one of the LUT entries of Sen2Cor. That is, for each specific LUT entry in Sen2Cor, we would have the values, uncertainty, and error correlation between the atmospheric functions.

We will use this method to compare against the proposed MCM multivariate in Section III-B in order to flag the main strengths and limitations.

III. Results

A. MCM Results Over Two Scenes

This section exemplifies a run and output of the software over two selected sites. These sites are located at the latitude and longitude −6.2869°, −66.8652° and 28.55°, 23.39°. They represent a typical Amazon forest and dune area of the Sahara desert. The latter is largely used by the community for radiometric validation of optical instruments onboard satellites [46]. The area selected for both sites is 500 × 500 m^2. Fig. 2 contains an RGB image for each one of the sites.

The error distributions for the TOA radiance result in a near-normal distribution as expected [6]. The relative uncertainty varies from 1% to 3% mostly depending on the radiance level. Fig. 3 contains the error distribution for a subset of bands of the surface reflectance that we have obtained by running the multivariate MCM code with 4000 samples. The mean and standard deviation for all the S2 L2A bands are included in Table I.

The relative uncertainty here varies largely depending on the band. The larger relative uncertainty occurs for B9 due to the large sensitivity to water vapor. The lowest relative uncertainty (in the range of 3%–5%) occurs for the rest of the bands in the NIR and SWIR. However, these relative values increase for those bands that are more sensitive to the atmosphere (e.g., B1–B4). In those bands, the values can range from 5% to values over 20%. The error distributions at surface reflectance follow a normal distribution for several bands. This situation does not occur for B9 and B1–B4 of the Amazon forest example. The nonnormal distribution of B9 is the result of a large impact of the water vapor absorption and its nonlinear response. The slight skewness in the B1–B4 distribution is due to the higher impact of the atmosphere.

Fig. 3 shows the result of combining different atmospheric functions and the additional correction steps (see Section II-B). These atmospheric functions are presented in Fig. 4 for the bands B1 and B9.

The distribution of the different atmospheric functions is close to normal for the desert example, supporting the surface reflectance results. In contrast, the distribution for the different atmospheric functions in the Amazon forest case is highly nonnormal and irregular. For some functions but mainly for the downwelling irradiance E_g, the distribution presents some clipping. This can be attributed to the input AOT distribution that, for the Amazon forest, is set to a mean of 0.07 and standard deviation of 0.68 [see (6)]. Thus, part of the input AOT has been clipped to zero. On the contrary, the AOT for Libya-4 is 0.21 with a standard deviation of 0.1, which results in a small number of clipped values. In general, larger clipping occurs for AOT values below 0.1 where the systematic uncertainty represents a floor value that rapidly scales in relative terms. This is realistic (i.e., a certain part of the uncertainty budget does not scale with the AOT value) but represents a challenge in the distribution modeling.

Despite larger irregular patterns in the distribution of the atmospheric functions, the results in the distribution at the surface reflectance only partially reflect a small nonnormal response. That is, to some extent, due to the combination of the atmospheric functions these effects are canceled out. For example, the downwelling direct (E_g^{dir}) and diffuse (E_g^{diff}) are completely inversely correlated (i.e., opposite distribution shapes).

In addition to the information on the specific per-band uncertainty, the software also generates the error correlation matrix between the S2 bands. The result for each one of the examples is given in Fig. 5.

At first glance, it is clear that the spectral error correlation is very different for the two sites. The Amazon example shows a strong correlation between B1–B4 bands and B6–B8A bands separately. These bands correspond to those dominated by the atmosphere and the surface signal, respectively. The error correlation for the desert case is significantly higher as a consequence of much higher surface reflectance. The correlation tends to decrease with the spectral distance except in the case of B9. In that case, the band is uncorrelated with most of the bands except for those that have a minor sensitivity to water vapor.

B. LPU Versus MCM Multivariate Approach

Section II-D described the methodology to obtain uncertainty estimates with an LPU methodology for the combination of the atmospheric functions. It was anticipated that the expected complexity of using this approach is due to the nonlinear response and ad hoc solutions of the radiative transfer code.

This section implements the same examples as in Section III-A but using both the LPU framework and MCM multivariate approach. The comparison only considers the propagation of the RTM input uncertainty and L1C uncertainty (see Fig. 1), that is, the combination described by (2). Further uncertainty sources related to the adjacency correction, the Lambertain assumption of the correction model, and the estimated accuracy of the LibRadTran software are not included. These have been modeled as a normal distribution in the absence of further information. Thus, not including these sources, facilitates the identification of disagreements in the intercomparison.

Table II presents the surface reflectance uncertainty differences between the LPUC and MCM combination for all L2A bands.

There is a high level of agreement for all bands except for those most affected by AOT (B1, B2, and B3) and water vapor (B9) where disagreements between an LPU approach and MCM multivariate appear.

We have produced another example analog to the one presented here, but Libradtran has been parameterized with a water vapor level of 5 cm and AOT of 1.14 (approximately 5-km visibility). This represents an extreme but realistic scenario since they correspond to the highest values stored in the precompiled LUT of Sen2Cor. Fig. 6 presents the surface reflectance distribution for a set of L2A bands.

The distribution for almost all the cases is highly nonnormal and disagrees with an LPU framework. Thus, these examples indicate that the LPU approach is not valid when the atmosphere dominates over the surface signal. Fig. 6(a) has a positive mean value of 0.039 but includes a fraction of the distribution in negative values as a result of an overcorrection of the atmosphere.

We have further extended the intercomparison to map LPU and MCM multivariate results as a function of both AOT and WV values. Fig. 7 presents the uncertainty map as a function of AOT and WV values for the Amazon site [see Fig. 2(a)] and B8. The calculation points for AOT are from 0 to 1 in 0.25 steps and WV gridding from 0 to 6 cm in 1.5-cm steps. A version in a reduced range is also calculated with AOT values from 0 to 0.3 in 0.05 steps and WV from 0 to 3 cm in 0.5-cm steps. The MCM uncertainty has been calculated with

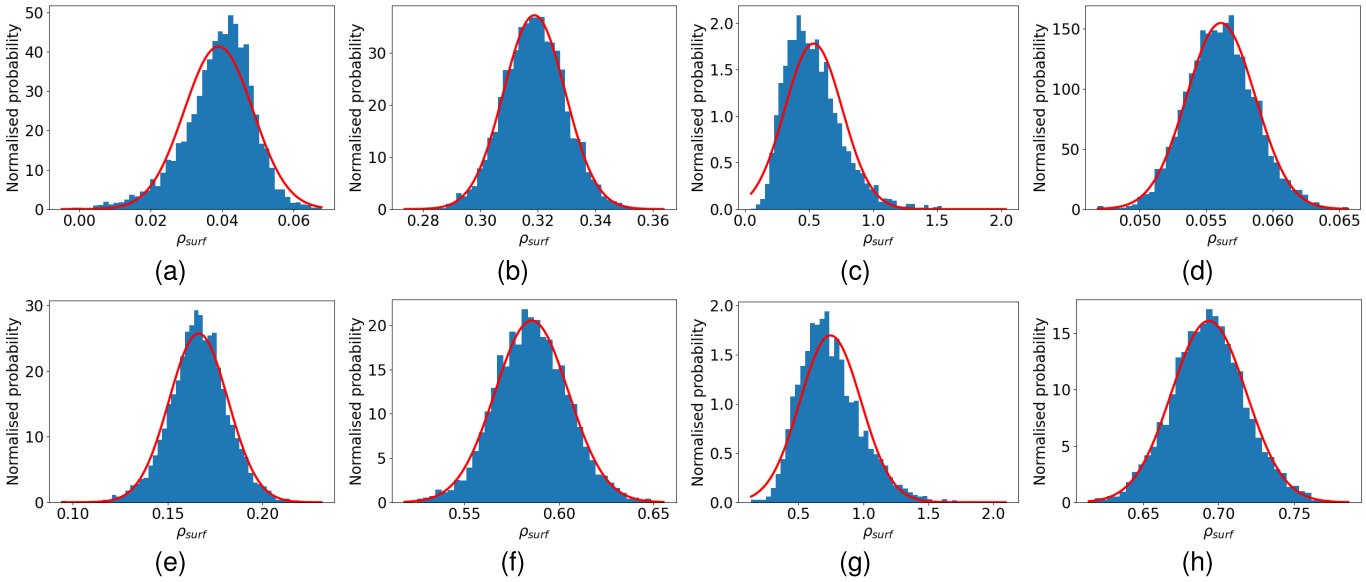


Fig. 3. Surface reflectance distribution obtained with the L2A-RUT code for bands B1, B8, B9, and B12 and the Amazon and Libya-4 sites described in Fig. 2. The red line corresponds to the ideal normal fitting. (a) Amazon B1. (b) Amazon B8. (c) Amazon B9. (d) Amazon B12. (e) Libya-4 B1. (f) Libya-4 B8. (g) Libya-4 B9. (h) Libya-4 B12.

TABLE I

SURFACE REFLECTANCE MEAN AND STANDARD DEVIATION OF THE S2 L2A BANDS FOR AMAZON AND LIBYA-4 CASES

Amazon	B1	B2	B3	B4	B5	B6	B7	B8	B8A	B9	B11	B12
Mean	0.039	0.038	0.050	0.029	0.078	0.232	0.297	0.319	0.341	0.535	0.148	0.056
Std[%]	24.72	19.39	10.85	13.45	5.70	3.79	3.45	3.36	3.39	41.82	3.92	4.60
Libya-4	B1	B2	B3	B4	B5	B6	B7	B8	B8A	B9	B11	B12
Mean	0.166	0.227	0.341	0.483	0.525	0.535	0.564	0.586	0.590	0.747	0.734	0.694
Std[%]	9.32	5.95	4.13	3.49	3.77	3.56	3.35	3.32	3.35	31.5	3.55	3.57

TABLE II

L2A UNCERTAINTY FOR AMAZON AND LIBYA-4 CASES USING AN MCM APPROACH AND LPU METHODOLOGY FOR THE COMBINATION OF THE ATMOSPHERIC FUNCTIONS. THE ADJACENCY CORRECTION, THE LAMBERTAIN ASSUMPTION OF THE CORRECTION MODEL, AND THE ESTIMATED ACCURACY OF THE LIBRADTRAN SOFTWARE ARE NOT INCLUDED

Amazon	B1	B2	B3	B4	B5	B6	B7	B8	B8A	B9	B11	B12
Std LPU[%]	12.61	11.20	6.59	8.72	3.67	1.93	1.47	1.34	1.67	31.07	1.91	3.13
Std MCM[%]	10.81	9.75	5.34	7.77	3.38	1.91	1.19	1.12	1.50	42.51	1.89	3.30
Libya-4	B1	B2	B3	B4	B5	B6	B7	B8	B8A	B9	B11	B12
Std LPU[%]	3.85	2.79	1.90	1.45	1.72	1.52	1.30	1.21	1.43	23.78	1.19	1.48
Std MCM[%]	1.56	1.46	1.26	1.24	1.97	1.57	1.22	1.16	1.38	30.60	1.26	1.71

500 samples for each run and two different methodologies: first as a standard deviation of the error distribution (MCM std.) and then as the 68.27% of probability around the median value (MCM unc.).

The results for B8 show generally large differences. These differences are significant for large values of WV where the LPU tends to overestimate the uncertainty. Observing the surface reflectance distribution in Fig. 6(b), it is visible that the small sensitivity of B8 to water vapor absorption results in a highly peaked distribution that leads to an overestimation of the LPU approach. There are also small differences between the MCM std. and MCM unc. The small overestimation of the MCM std. with high water vapor that can be associated with the right skewness of the distribution in Fig. 6(b).

IV. DISCUSSION

A. Main Findings

Section III-A presents two examples (one for the Amazon forest and the other for the Sahara dune desert) that were useful to test the code and examine the first results.

The first interesting point is that in comparison to the L1C uncertainty estimates, the L2A uncertainty estimates are larger in relative terms and present a larger spectral band dependence. That is, the range of relative uncertainty at L1C moves from 1% to 3% for the examples and 3%–20% for the L2A uncertainty estimates. The atmospheric correction increases the relative uncertainty by reducing the available signal and introducing an uncertainty associated with the

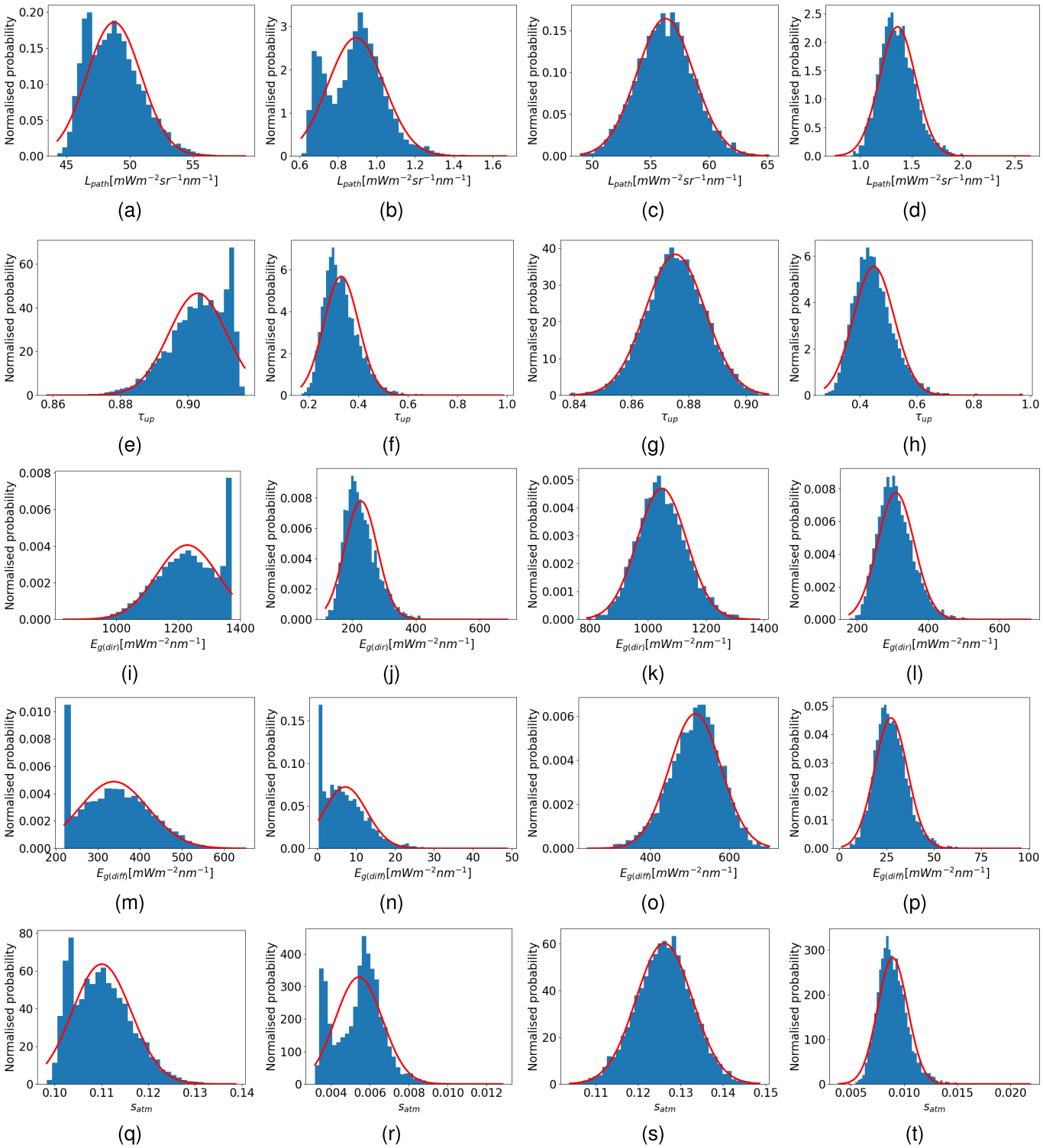


Fig. 4. Atmospheric function distribution (L_p, τ, E_g^{dir}, E_g^{diff}, and s) obtained with the L2A-RUT code for bands B1 and B9 and the Amazon and Libya-4 sites described in Fig. 2. The red line corresponds to the ideal normal fitting. (a) Amazon B1 L_p. (b) Amazon B9 L_p. (c) Libya-4 B1 L_p. (d) Libya-4 B9 L_p. (e) Amazon B1 τ. (f) Amazon B9 τ. (g) Libya-4 B1 τ. (h) Libya-4 B9 τ. (i) Amazon B1 E_g^{dir}. (j) Amazon B9 E_g^{dir}. (k) Libya-4 B1 E_g^{dir}. (l) Libya-4 B9 E_g^{dir}. (m) Amazon B1 E_g^{diff}. (n) Amazon B9 E_g^{diff}. (o) Libya-4 B1 E_g^{diff}. (p) Libya-4 B9 E_g^{diff}. (q) Amazon B1 s. (r) Amazon B9 s. (s) Libya-4 B1 s. (t) Libya-4 B9 s.

correction. For those bands with little impact on the atmosphere, the surface reflectance distribution is largely normal. However, this is not the case for those bands where the atmosphere dominates. For example, the error distribution for B9 is generally nonnormal due to its sensitive to water vapor.

The results in Section III-A illustrate important differences between the two sites not only at uncertainty levels but more

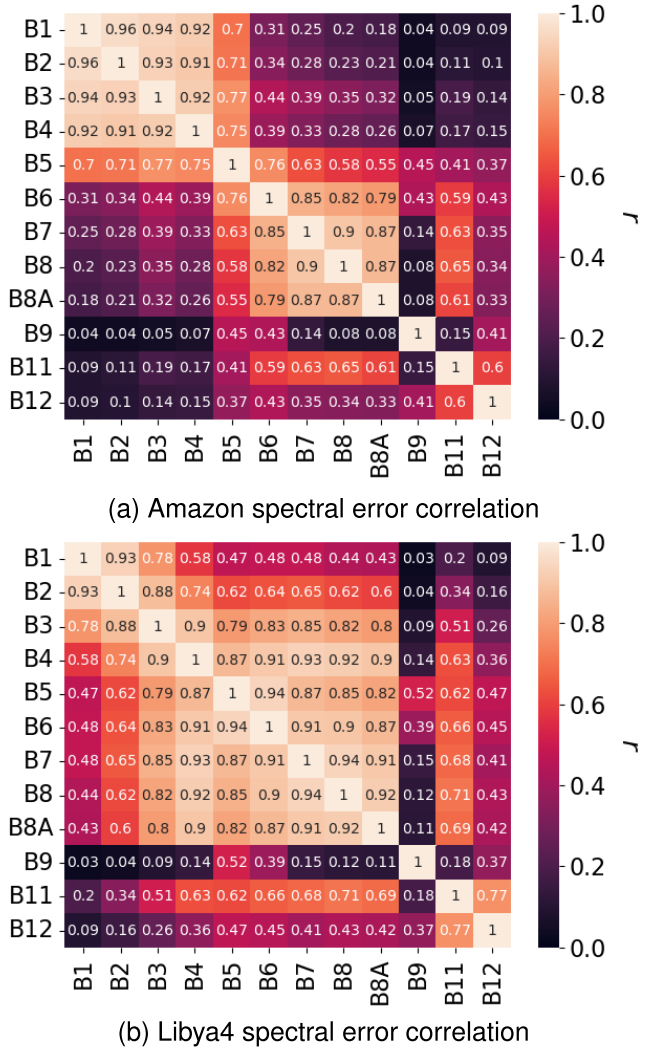


Fig. 5. S2 L2A spectral error correlation for (a) Amazon site and (b) desert site.

importantly on their spectral error correlation. The Amazon example shows a strong correlation between B1–B4 bands and B6–B8A bands separately as a consequence of the vegetation's red edge. The error correlation in the desert case is higher due to the higher surface reflectance with a dependence on the spectral distance.

We also explored in detail the nonnormal distribution of the surface reflectance and its implications for calculating the uncertainty using different methodologies. The individual atmospheric functions presented in Fig. 4 can individually show the nonnormal and irregular distribution of each one of them. On the one hand, it shows that the combination of some of these functions (e.g., diffuse and direct irradiance) is partially canceled when combined. On the other hand, it also shows the effect of clipping input distributions such as AOT. In that case, it shows that we need to consider more realistic and potentially nonnormal distributions of the input sources.

Compared to an LPU approach (see Section II-D), the study in Section III-B has illustrated the limits of assuming normal distributions and linear responses. Setting extreme but possible atmospheric conditions with a water vapor level of 5 cm and AOT of 1.14 (approximately 5-km visibility), the results between the two methods largely disagree for all bands. This comparison was refined by producing an uncertainty map as a function of AOT and WV in Fig. 7. It shows that the LPU tends to overestimate the uncertainty with an important effect of water vapor in B8. There were also differences between the standard deviation of the output distribution and the direct assessment of the 68.27% probability around the median.

In general, the LPU method is valid as long as the atmosphere is not dominant and input sources are normally distributed. Any band that is highly sensitive to atmospheric absorption and scattering will be affected. For the S2 satellite mission, this occurs for some bands and scenes. That is, the proposed methodology proposed in Section II-D has a limited application at a global scale. Upcoming Copernicus satellite missions such as CHIME or S2-NG will include a larger and narrower number of bands. It is expected that a more detailed and reviewed multivariate MCM model is needed to assess the uncertainty and its comparison against other alternative approaches.

B. Further Work

Although a large number of uncertainty sources are included, some of them are not yet available in this version. Under some scenarios, it is possible that this results in a small underestimation of the current L2A uncertainty levels. However, it is the idea that subsequent versions of this software include refined and/or novel uncertainty sources.

The work in [6] describes the missing contributions for the L1C uncertainty estimates. These are effects such as spectral response, polarization, or orthorectification that depends on the scene. Relevant improvements are ongoing to model these contributions. For example, the work in [47] describes the efforts to provide in the upcoming S2 satellite units (C/D) a per-pixel spectral response. The orthorectification of the S2 data products at L1C (TOA reflectance) has been also studied in [48].

The work here presented includes the uncertainty associated with the atmospheric correction that is applied to the S2 L2A data products. The contributions described in Section II-B include the most relevant effects of this process. However, there are specific corrections that are not considered here. Both the terrain and cirrus correction are included as an option in the Sen2Cor software and should be specifically studied in subsequent revisions. Sen2Cor also contains an LUT with precompiled values for the atmospheric functions at different parameterizations. Thus, in a real scenario, the atmospheric functions are interpolated values from the LUT and require the assessment of the interpolation error. The uncertainty associated with the atmospheric profile and AOT model is also an important and challenging contribution to be considered. For the latter, we exemplify its impact on L2A uncertainty estimates and discuss potential implementations in Section IV-C.

For the uncertainty sources that have been included in this first prototype version, we have included a simplified allocation value in several cases. More important is that these input distributions have been assumed Gaussian. The

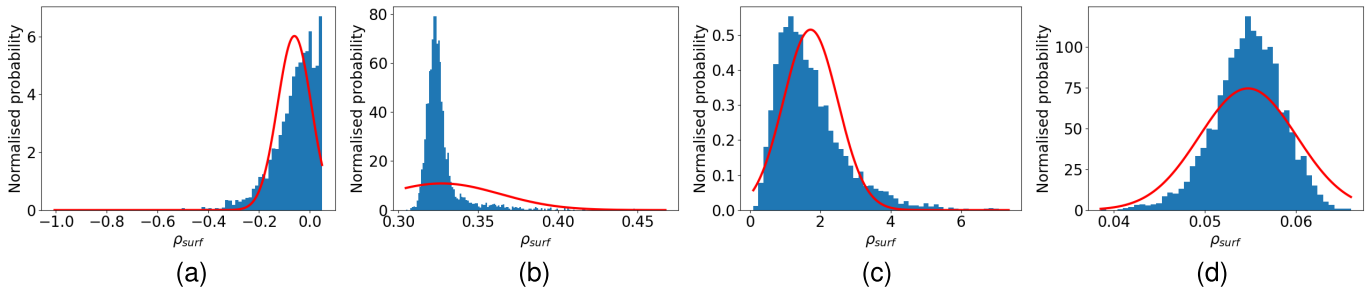


Fig. 6. Surface reflectance for a set of bands in the Amazon case obtained for a uniform Lambertian approach [i.e., (2)] for both the MCM multivariate and LPU approaches. The simulation was forced to set a value of 5 cm for water vapor and 1.14 (approximately 5-km visibility) for AOT. The red line represents the normal distribution associated with the uncertainty value of the LPU approach. (a) B1. (b) B8. (c) B9. (d) B12.

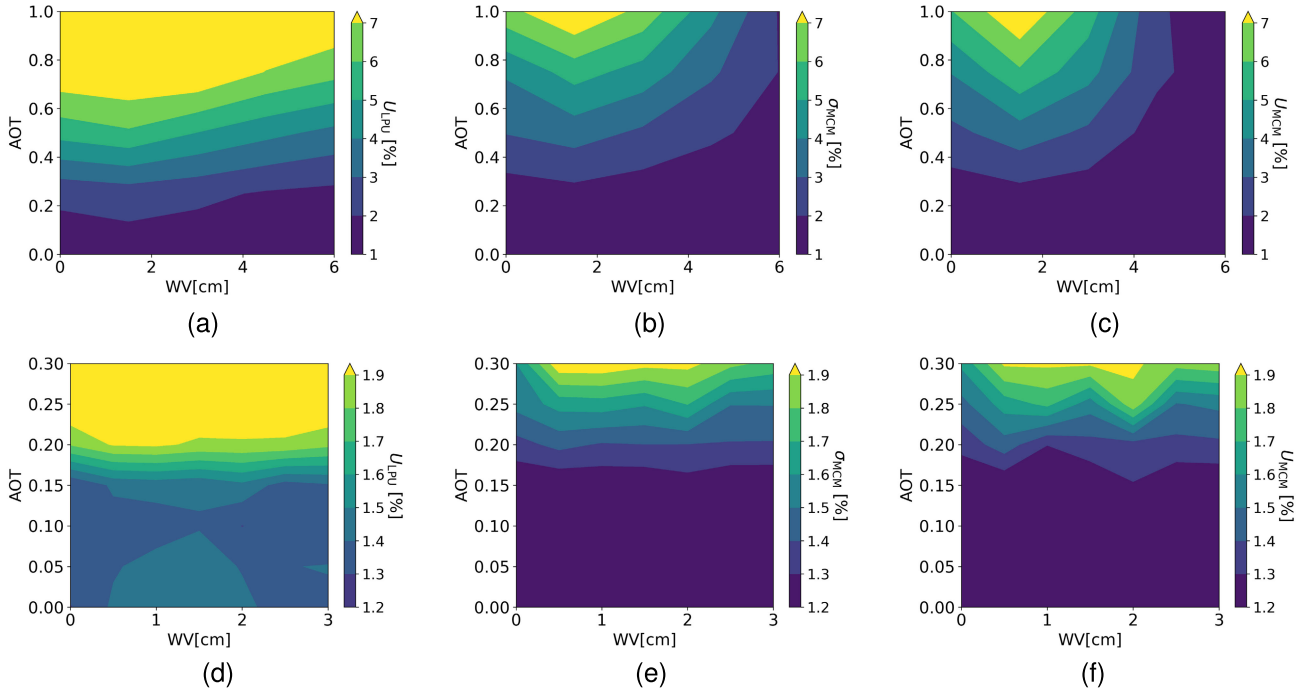


Fig. 7. Map of surface reflectance uncertainty as a function of AOT and WV in the Amazon case obtained for a uniform Lambertian approach [i.e., (2)] for the MCM multivariate and LPU approach and band B8. The former method has been calculated as a standard deviation of the error distribution (MCM std.) and the 68.27% probability around the median value (MCM unc.). The top three figures have been obtained from AOT gridding: 0–1 in 0.25 steps and WV gridding from 0 to 6 cm in 1.5-cm steps. The bottom three figures set an AOT gridding from 0 to 0.3 in 0.05 steps and WV from 0 to 3 cm in 0.5-cm steps. (a) B8 (LPU). (b) B8 (MCM std.). (c) B8 (MCM unc.). (d) B8 (LPU). (e) B8 (MCM std.) f) B8 (MCM unc.).

results in Section III-A show that this might be limited and unrealistic for parameters such as AOT and WV. Furthermore, the proposed uncertainty estimates are highly dominated by the systematic errors obtained during validation [31] (see Section II-B2). This is a continuous effort, and future updates might consider more detailed information such as a per-biome uncertainty. Other uncertainty sources such as the adjacency effect $u(\overline{\rho^1})$ are based on first guess values. In the absence of other information, this is a valid approach. However, subsequent studies should be dedicated to assess more specific uncertainty estimates for these sources.

Finally, one of the most important novelties in this study is the consideration of spectral error correlation through the entire processing chain. Uncertainty contributions here are assigned spectral error correlation matrices dependent on the spectral distance and the focal plane (VNIR or SWIR). Nonetheless, we are aware that the information available is limited, and contributions such as *Libradtran uncertainty* and *lambertian assumption* (see Sections II-B5 and II-B4, respectively) have a strong impact on the final budget. In general, the attachment of spectral error correlation information in preflight calibration certificates or atmospheric studies should be encouraged in future studies.

C. Considering the Uncertainty on AOT Models

Sen2Cor calculates the atmospheric functions based on the Libradtran default aerosol according to [49]. These properties refer to a rural-type aerosol in the boundary layer, background aerosol above 2 km, spring–summer conditions, and a visibility of 50 km. These optical or microphysical settings may be modified in Libradtran [21]. In the Sen2Cor user mode, the aerosol profile in the boundary layer can be modified, and it is automatically detected by the L2A-RUT code. However, no uncertainty is associated with the default profile expecting to introduce important errors where the AOT is high and profiles differ from the standard case.

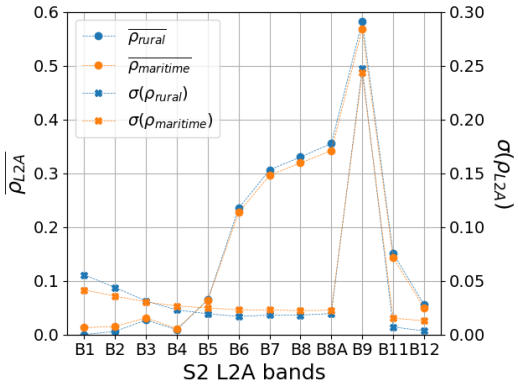


Fig. 8. Mean and standard deviation of the L2A surface reflectance distribution for the Amazon scene with an adjusted AOT value of 0.5 and considering the default rural boundary layer aerosol and maritime type one.

To exemplify the impact that the AOT model could have on the estimates, we run the L2A-RUT for the Amazon case with a value of AOT 0.5 with both the rural and maritime boundary layer aerosols as in [49]. The mean and standard deviation for all L2A bands are displayed in Fig. 8.

The results indicate important differences both in terms of mean and standard deviation with the maritime model showing higher mean values in the lower wavelengths and higher values at longer wavelengths (vice versa for the standard deviation). Future versions will require an important effort that is not so much focused on the possibility of modeling the AOT profiles (Libradtran and other RTC are well-suited for this purpose) but rather on associating uncertainty to a per-pixel case. Potential solutions that could be explored include the association of a potential range of values over specific sites based on aerosol properties climatology (e.g., [50]).

D. Toward an Operational Version

The current implementation in [1] allows the users to a rapid access to queries for an uncertainty and spectral error correlation assessment of a pixel or set of them. It is an ideal solution for small-area studies or a general understanding of the expected L2A uncertainty and spectral correlation over scene types. However, the proposed multivariate MCM is computationally expensive, and the amount of information at a pixel level becomes prohibitive. Thus, this methodology is not expected to run at an operational level but serve as the basis to explore and validate different strategies for operational implementation.

Using an LPU method is not discarded here, but we have seen that its domain of validity is reduced compared to a similar MCM model. Thus, it could be possible to set up the LPU method but restrict its applicability to valid cases or modify it to include higher order derivatives. Otherwise, consideration should be given to using MCM or alternative approaches [51]. In a way, the implementation of an MCM model as here is recommended as a validation tool for an LPU or other methodology.

The calculation of the uncertainty and spectral error correlation for a single pixel, area, or set of pixels does not constitute a major challenge in terms of data volumes. However, the per-pixel calculation and storage of these data volumes might become unfeasible. Upcoming Copernicus satellite missions such as CHIME or S2-NG [52] will further increase the demand with a larger set of bands and higher spatial resolution. Thus, alternatives must be sought for a global distribution of this information.

In recent years, it has been common practice to consider emulators as fast approximations of the atmospheric correction model. We can find already examples such as the one in [53] where they achieved an emulator with an accuracy $<1\%$ ($k = 2$) with respect to MODTRAN radiative transfer. Another example in [54] proposes a surrogate model to generate a candidate result that is refined with an emulator that corrects for the difference between the surrogate model and a complex accurate RTM. Translating this from atmospheric correction into surface reflectance uncertainty, it could be possible to train such models with an MCM model similar to the one presented here. Thus, these types of models could be instrumental in delivering not only uncertainty but also spectral error correlation or error distributions at a pixel level to the users.

An alternative atmospheric correction method is to consider the mapping between TOA observations and surface reflectance with Bayesian methods, such as the optimal estimation method [55]. For example, the work in [56] implemented an atmospheric correction that can be applied to generate S2 L2A products. Compared to a multivariate MCM and an LPU approach (see Sections II-C and II-D), this method has the advantage to directly offer the variance matrix (both uncertainty and spectral error correlation matrixes) in the retrieved space. However, the uncertainty propagation in optimal estimation relies on prior data that do not necessarily approximate the real scene such as in the case of extreme events. Furthermore, it also assumes a normal distribution, unbiased measurements, and a moderately linear Jacobian matrix [18].

Thus, a similar cross-comparison as in Section III-B can be applied to, for example, identify scenarios where the Jacobian of the atmospheric model might not be sufficiently linear. Indeed, it is possible not only to identify but also to adapt these Bayesian frameworks so that these limitations are (partially) overcome. For example, the work in [57] also defines a multivariate MCM sampling of the Bayesian retrieval algorithm so that the retrieved covariance matrix accounts for the model discrepancy.

V. Conclusion

We presented the L2A-RUT software tool (available at [1]) that delivers both uncertainty estimates and spectral error correlation from the S2 L2A data products (i.e., surface reflectance). The methodology considers the uncertainty of the S2 L1C data products and the atmospheric correction. We have set an MCM multivariate model that accounts for the spectral error correlation between S2 L2A bands and propagates the uncertainty of the L1 TOA reflectance, atmospheric parameterization, and adjacency correction. On top of this propagation, we also model the contributions from the Lambertain assumption and the estimated accuracy of the atmospheric radiative transfer. The results over a forest and desert scene illustrate the large variations both in uncertainty levels and spectral error correlation that can be found between data products. The selection of the MCM multivariate model over an LPU

scheme is justified because the error distribution cannot be systematically described as a normal distribution, the transformation can be nonlinear, and the atmospheric correction is not fully described by a mathematical model. An intercomparison between these two combination models has shown important differences that are dependent on the spectral band, AOT, and WV levels. In addition, we observe how the modeling of input distributions (e.g., AOT) as a normal distribution can be limited and unrealistic in error propagation. The current tool version here presented delivers information to the user for a selected pixel, small area studies, or a general estimation of the L2A uncertainty and spectral correlation over scene types. The multivariate MCM here proposed cannot be extended to process global per-pixel and disseminate both L2A uncertainty and error spectral correlation. Nonetheless, it can be helpful to validate and study different implementation strategies at an operational level so that per-pixel uncertainty and spectral error correlation can be delivered to the users.

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